

Which Continent is Happiest? An analysis of Happiness Scores and modeled contributing factors across continents

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Introduction

Happiness is a key component of the human experience, and it is shaped by economic, social, and health-related factors. The consistently high rankings of Nordic countries in the World Happiness Report suggest that happiness may follow geographical patterns. However, it remains unclear whether such patterns are also apparent on a larger geographical scale, such as between continents. Additionally, this raises the question of whether differences in Happiness Scores are accompanied by different patterns of modeled factor contributions across continents. Therefore, this report investigates the following research question: Do average Happiness Scores differ across continents, and are there differences in the factors contributing to these scores?

Data sample

The dataset [1] used in this report is the 2019 Happiness Report Dataset, which was retrieved via Kaggle. It contains data on the Happiness Scores of 156 countries, which originate from the Gallup World Poll. Additionally, for each country, the dataset contains data on the modeled contribution of GDP per capita, social support, healthy life expectancy, freedom to make life choices, generosity, and perceptions of corruption to the Happiness Score. Quality checks demonstrated that there were no missing values or duplicates, so no data cleaning was neccessary. To group country data into continents, the pycountry-convert package from python was used.

For the analysis, the average Happiness Scores and standard deviation were calculated and visualized in a bar chart with error bars. A Welch's ANOVA test was conducted to test for statistical significance. Additionally, average modeled contribution for each of the six factors was calculated for every continent and visualized in a second bar chart to compare relative contributions.

Results and discussion

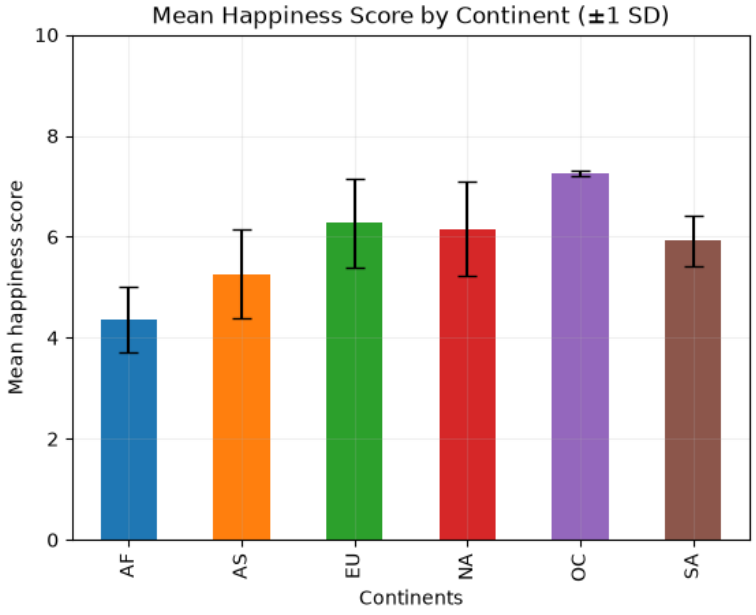


Figure 1: Mean Happiness Scores by continent, with error bars representing one standard deviation. Based on the 2019 World Happiness Report dataset.

The average happiness scores differ across continents, as shown in Figure 1. Oceania has the highest average Happiness Score (7.268) and Africa has the lowest (4.368). The error bars indicate that there is

considerable variation within the continents. However, since the dataset includes only two countries from Oceania, its average may not be representative of the continent, which also explains the small standard deviation. A Welch's ANOVA test, excluding Oceania due to this limitation, resulted in a p-value < 0.01 ($p = 1.06 \times 10^{-13}$), indicating that the differences are statistically significant.

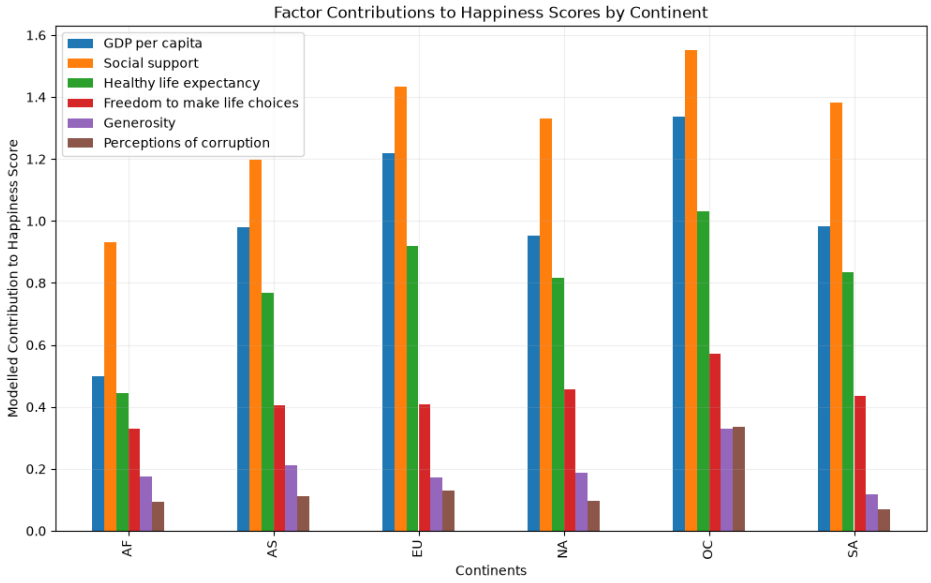


Figure 2: Average modelled factor contributions to Happiness Scores across continents.Based on the 2019 Happiness Report dataset.

The average modeled factors follow a similar pattern across continents as shown in Figure 2. Social support has the largest average modeled contribution in every continent, followed by GDP per capita, and healthy life expectancy. Differences lie, however, in the magnitude of each contribution.

Overall, the results suggest that there are geographical differences in average happiness across continents. These differences might be associated with differing strengths of economies, social systems, or health systems. Despite differences in Happiness Scores, the relative importance of the modeled factor contributions appears similar across continents, with social support consistently making the largest average contribution. Several limitations should be considered. First, as already mentioned, Oceania is represented by only two countries, which makes its results less representative. Second, the data may not fully satisfy the assumptions of the Welch's ANOVA test.

References

[1] Sustainable Development Solutions Network (2019) *World Happiness Report*. Available at: https://www.kaggle.com/datasets/unsdsn/world-happiness.